

Galois Theory

Solution to Selected Exercises and Problems

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Chapter 1

Exercise 5

Let $S = \{p + q\alpha + r\alpha^2 \mid p, q, r \in \mathbb{Q}\}$. We first prove that S is a subring of $\mathbb{C}$.

Note that $1 = 1 + 0\alpha + 0\alpha^2 \in S$. Let $p + q\alpha + r\alpha^2, p' + q'\alpha + r'\alpha^2 \in S$, where $p, q, r, p', q', r' \in \mathbb{Q}$. Then we have $(p + q\alpha + r\alpha^2)(p' + q'\alpha + r'\alpha^2) = (p + p') + (q + q')\alpha + (r + r')\alpha^2 \in S$, $-(p + q\alpha + r\alpha^2) = (-p) + (-q)\alpha + (-r)\alpha^2 \in S$ and $(p + q\alpha + r\alpha^2)(p' + q'\alpha + r'\alpha^2) = (pp' + 2rq') + (pq' + p'q + 2rr')\alpha + (pr' + qq' + qr' + rp')\alpha^2 \in S$. By the subring test, S is a ring.

Next, we construct an inverse explicitly to show that S is a subfield. Let $x = p + q\alpha + r\alpha^2 \in S \setminus \{0\}$, where $p, q, r \in \mathbb{Q}$. Then we have $x^2 = (p^2 + 4qr) + (2pq + 2r^2)\alpha + (2pr + q^2)\alpha^2$ and $x^3 = (p^3 + 12pqr + 2q^3 + 4r^3) + (3p^2q + 6pr^2 + 6q^2r)\alpha + (3p^2r + 3pq^2 + 6qr^2)\alpha^2$. Let $x^3 + k_2x^2 + k_1x + k_0 = 0$. Denote the coefficient of α^j in x^i by x_{ij} . Then we have $(x_{30} + x_{31}\alpha + x_{32}\alpha^2) + k_2(x_{20} + x_{21}\alpha + x_{22}\alpha^2) + k_1(p + q\alpha + r\alpha^2) + k_0 = 0$, that is, $(x_{30} + x_{20}k_2 + pk_1 + k_0) + (x_{31} + x_{21}k_2 + qk_1)\alpha + (x_{32} + x_{22}k_2 + rk_1)\alpha^2 = 0$. Since $1, \alpha, \alpha^2$ are linearly independent, comparing coefficients gives us the linear system

$$\begin{cases} k_0 + pk_1 + x_{20}k_2 = -x_{30} \\ qk_1 + x_{21}k_2 = -x_{31} \\ rk_1 + x_{22}k_2 = -x_{32} \end{cases}.$$

We can use the following python code to solve this system.

```

1  import sympy as sp
2
3  k0, k1, k2 = sp.symbols('k0 k1 k2')
4  p, q, r = sp.symbols('p q r')
5
6  x30 = p**3 + 12 * p * q * r + 2 * (q**3) + 4 * (r**3)
7  x31 = 3 * (p**2) * q + 6 * p * (r**2) + 6 * (q**2) * r
8  x32 = 3 * (p**2) * r + 3 * p * (q**2) + 6 * q * (r**2)
9  x20 = p**2 + 4 * q * r
10 x21 = 2 * p * q + 2 * (r**2)
11 x22 = 2 * p * r + q**2
12
13 M = sp.Matrix([
14     [1, p, x20, - x30],
15     [0, q, x21, - x31],
16     [0, r, x22, - x32]
17 ])
18
19 solution = sp.solve_linear_system(M, k0, k1, k2)
20
21 for var in [k0, k1, k2]:
22     print(f"{var} =")
23     sp.pprint(sp.simplify(solution[var]))

```

The solution is $k_0 = -p^3 + 6pqr - 2q^3 - 4r^3$, $k_1 = 3p^2 - 6qr$ and $k_2 = -3p$. Since $x^3 + k_2x^2 + k_1x + k_0 = 0$, we have $x(x^2 + k_2x + k_1 + k_0x^{-1}) = 0$. Recall that $\mathbb{C}$ is a field and hence a domain, and $x \neq 0$. Therefore, $x^2 + k_2x + k_1 + k_0x^{-1} = 0$, that is,

$$x^{-1} = -\frac{1}{k_0}x^2 - \frac{k_2}{k_0}x - \frac{k_1}{k_0}.$$

Since x and x^2 are linear combinations of $1, \alpha, \alpha^2$, x^{-1} is also a linear combination of $1, \alpha, \alpha^2$. □